

Stress-Driven 3D-IC Placement with TSV Keep-Out Zone and Regularity Study

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Abstract—Through-silicon via (TSV) fabrication causes tensile stress around TSVs which results in significant carrier mobility variation in the devices in their neighborhood. Keep-out zone (KOZ) is a conservative way to prevent any devices/cells from being impacted by the TSV-induced stress. However, owing to already large TSV size, large KOZ can significantly reduce the placement area available for cells, thus requiring larger dies which negate improvement in wirelength and timing due to 3D integration. In this paper, we study the impact of KOZ dimension on stress, carrier mobility variation, area, wirelength, and performance of 3D ICs. We demonstrate that, instead of requiring large KOZ, 3D-IC placers must *exploit* TSV stress-induced carrier mobility variation to *improve* the timing and area objectives during placement. We propose a new TSV stress-driven force-directed 3D placement that consistently provides placement result with, on average, 21.6% better worst negative slack (WNS) and 28.0% better total negative slack (TNS) than wirelength-driven placement.

I. INTRODUCTION

Three-dimensional (3D) chip stacking has emerged as a viable technology to continue the trend of ever increasing integration and functionality of electronic devices predicted by Moore's law. Such technology has become feasible due to the advent of through-silicon vias (TSV), which allow electrical connection between functional modules across multiple dies inside a single 3D chip stack. TSVs, however, are much larger than all other standard cells in a design, and thus impact various figure of merits of 3D integrated circuits (3D IC) such as wirelength, area, and performance. A study on the physical impact of TSVs on 3D-IC layout, such as footprint and wirelength, was presented in [1]; however, the work did not consider keep-out zone (KOZ) surrounding TSVs.

TSV fabrication causes tensile mechanical stress around TSVs because of the mismatch in the coefficients of thermal expansion (CTE) between silicon ($3 \times 10^{-6} K^{-1}$) and copper ($17 \times 10^{-6} K^{-1}$), a widely used material for TSV fill [2]. After cooling down to room temperature, copper contracts much faster than, and pulls the surface of surrounding silicon, causing tensile stress in the area [3]. Severe stress can result in cracking and damage in substrate and devices on top [4]. Moreover, stress causes hole and electron mobility variation in

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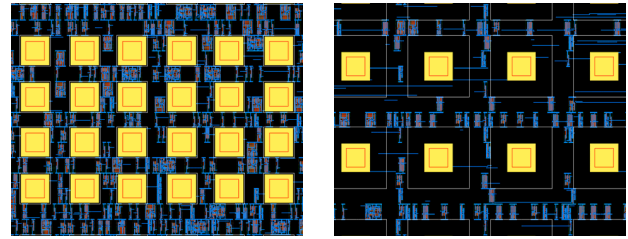


Fig. 1. Layouts with small versus large KOZ around TSVs. TSV landing pads are large yellow squares.

devices, which can result in performance degradation without proper control. KOZ is the area surrounding each TSV from which all logic cells must “keep out” so that they are not influenced by the TSV-induced stress.

To determine the size of KOZ in [5], the magnitude of stress caused by TSVs was studied and analyzed. KOZ is usually large because it is defined such that stress outside it is under preset tolerance. In real designs, the presence of abundant TSVs in use already has tremendous impact on 3D-IC layout. Large KOZ only worsens the situation as illustrated in Fig. 1 as it reduces the TSV stress-induced carrier mobility variation in surrounding logic cells at the cost of increasing die size.

To reduce KOZ without adverse electrical effect, we propose that placers must also consider the effect of TSV-induced stress on carrier mobility variation. Logic cells on critical paths must be placed in the position where the carrier mobility inside their PMOS/NMOS is not degraded (if not enhanced) by TSV-induced stress. Engineered stress has been widely used in industry to improve chip performance [6]. Few academic works also proposed placement perturbation techniques to use shallow trench isolation-induced stress [7] and strained silicon [8] for the same purpose. A recently developed compact TSV-induced stress and mobility variation model and stress-aware 3D static timing analysis (SA 3D STA) [9] can be used to guide the placers. Therefore, the necessity to keep large KOZ for electrical reason starts becoming obsolete.

The contributions of this work are as follows:

- We present the first placement algorithm to exploit hole and electron mobility variation caused by TSV-induced stress. A TSV-induced stress-driven force-directed 3D global placement is proposed in this paper. We introduce carrier mobility-based forces to a 3D force-directed quadratic placer, and describe how to balance them

against original placement forces.

- We devise a way to integrate our placer to commercial tools. The design flow enables us to perform, on GDSII-level 3D layouts, trial or detail route, parasitic extraction, and finally SA 3D STA to find accurate critical paths and critical nets/cells on them to guide the placer.
- Using the above mentioned design flow, we study the impact of KOZ on stress, carrier mobility variation, area, wirelength, and performance of 3D ICs. To the best of our knowledge, this is the first work that studies TSV-induced stress and carrier mobility variation issues in the context of KOZ and 3D-IC placement optimization.

The rest of this paper is organized as follows. First, the compact TSV-induced stress and mobility variation models and SA 3D STA [9] are reviewed in Section II. We describe our TSV-induced stress-driven design flow in Section III. In Section IV, we present our TSV-induced stress-driven force-directed 3D global placement. Experimental results are shown in Section V, followed by conclusions in Section VI.

II. PRELIMINARY WORKS

A. Mobility Variation Modeling

To take TSV-induced stress into account for SA 3D STA, an analytical model of TSV stress-induced carrier mobility variation was proposed in [9]. Carrier mobility change depends on not only applied stress, but also orientation between the stress and a transistor channel. The effect from multiple TSVs can be combined by using linear super-position. Total carrier mobility variation $\Delta\mu/\mu_{\text{total}}$ is defined in [9] as

$$\frac{\Delta\mu}{\mu_{\text{total}}} = \sum \frac{\Delta\mu}{\mu}(\theta) = -\Pi \sum_{i \in \text{TSVs}} (\sigma_i \cdot \alpha(\theta_i)), \quad (1)$$

where Π is the tensor of piezo-resistive coefficients, σ_i is the tensile stress caused by i^{th} TSV, $\alpha(\theta_i)$ is the orientation factor of i^{th} TSV, and θ_i is the degree between the center of the i^{th} TSV and a point at which we want to obtain mobility variation.

B. Stress-Aware 3D Static Timing Analysis

Even though the layout of a cell is fixed, its timing characteristic can vary based on TSV stress-induced carrier mobility variation. The SA 3D STA framework in [9] renames cells in verilog netlist to reflect their carrier mobility variation. For example, INVX1_N8_P8 is INVX1 with negative 8% hole mobility change and positive 8% electron mobility change. Cells with different mobility corners are characterized to make carrier-mobility variation-aware library.

In the framework, a verilog netlist and a parasitic extraction file (SPEF) for each die is prepared. A top level verilog netlist instantiates the dies, and connects them using wires, which correspond to TSVs. Finally, with a top level SPEF file for the TSVs, PrimeTime can provide the SA 3D STA results.

III. TSV-INDUCED STRESS-DRIVEN PLACEMENT OPTIMIZATION

In this section, we present an overview of our stress-driven timing optimization methodology. Basically, we use

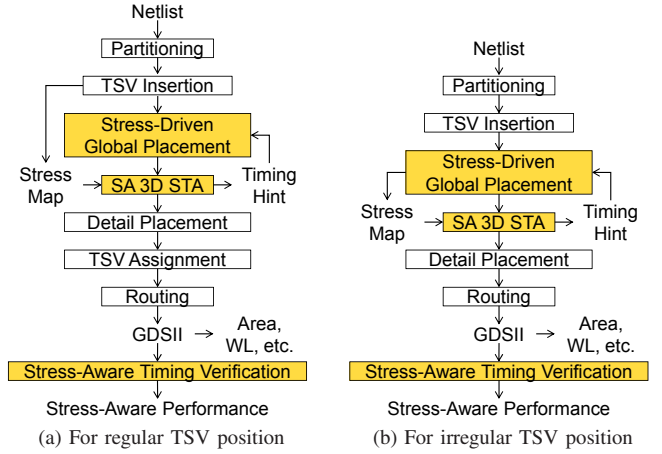


Fig. 2. Design flow for TSV-induced stress-driven placement optimization.

the placement styles in [1] and develop our 3D placer for stress-driven timing optimization as shown in Fig. 2. The framework in [1] supports two different TSV placements, namely, regular and irregular TSV position. In the case of regular TSV position, TSVs are placed at regular grid-like sites over the die area, and any net that needs to span multiple dies must connect to these TSVs. In the case of irregular TSV position, TSV and logic cell locations are determined simultaneously. We modify global placement stage for stress-driven timing optimization because we have flexibility to move cells to improve stress-aware timing. We do not perform any kind of stress-driven timing optimization during routing stage because TSV-induced stress mainly affects cell delay based on cell location, which is not changed during routing.

For design with regular TSV position, called TSV-site in [1], we start by partitioning logic cells into dies of a 3D chip stack using a min-cut approach. Then, we estimate the minimum number of signal TSVs required, and pre-place them on the dies. Knowing location of pre-placed signal TSVs, we calculate stress map on all dies for use during SA 3D STA. Then, we perform TSV stress-driven global placement, which is presented in Section IV, to obtain placement result. Note that our placer calls SA 3D STA in order to obtain the sets of nets and cells on critical paths to be optimized after every predefined iteration. Then, we perform detail placement, and assign TSVs to multiple-die nets in the 3D chip stack using the same method as in [1]. After routing, we can evaluate stress-aware performance from GDSII layout.

For design with irregular TSV position, called TSV co-placement in [1], the flow differs from that for design with regular TSV position in a few ways. After partitioning, TSVs are included into netlist as part of placement cells of multiple-die nets using the same heuristic, called net splitting, as in [1], and TSV assignment stage is not needed. Because TSV position is changed in every placement iteration, stress map needs to be regularly updated.

The presented design flow allows us to study the impact of KOZ on stress, carrier mobility variation, area, wirelength, and

performance of 3D ICs. The result of our study is analyzed, and reported in detail in Section V.

IV. TSV STRESS-DRIVEN GLOBAL PLACEMENT

In this section, we describe our TSV stress-driven global placement algorithm. It is based on a forced-directed quadratic placement [10], which was extended to support 3D-IC design in [1]. We introduce carrier mobility-based forces, and describe how to balance them against original placement forces in both works. We also discuss convergence of the algorithm when placing design with many TSVs with large KOZ.

A. Overview of 3D Force-Directed Quadratic Placement

Placement result in a quadratic placement [10] is computed by minimizing the quadratic wirelength function Γ defined as $\Gamma = \Gamma_x + \Gamma_y$, where Γ_x and Γ_y are wirelength along x- and y-axis. We can minimize Γ_x and Γ_y separately because they are independent. For brevity, only description for x-dimension is given in this paper because it similarly applies to y-dimension. Here, Γ_x can be written in a matrix form as

$$\Gamma_x = \frac{1}{2} \mathbf{x}^T \mathbf{C}_x \mathbf{x} + \mathbf{x}^T \mathbf{d}_x + \text{constant}, \quad (2)$$

where vector $\mathbf{x}$ represents the x-position of cells to be placed, matrix $\mathbf{C}_x$ represents the connection among the cells along x-axis, and vector $\mathbf{d}_x$ represents the connection to fixed pins along x-axis. Treating Γ_x as the energy of a spring system, we can view its derivative as net force $\mathbf{f}_x^{\text{net}}$ defined as

$$\mathbf{f}_x^{\text{net}} = \nabla_x \Gamma_x = \mathbf{C}_x \mathbf{x} + \mathbf{d}_x, \quad (3)$$

where ∇_x is vector differential operator. The minimum Γ_x is found when $\mathbf{f}_x^{\text{net}}$ is zero, but cells can overlap in few small areas. Density-based force $\mathbf{f}_x^{\text{den}}$ spreads cells away to reduce cell overlap, and is defined as

$$\mathbf{f}_x^{\text{den}} = \mathbf{C}_x^{\text{d}} (\mathbf{x} - \hat{\mathbf{x}}^{\text{d}}), \quad (4)$$

where vector $\hat{\mathbf{x}}^{\text{d}}$ represents the x-position of target points to which cells are connected by density-based springs, and diagonal matrix $\mathbf{C}_x^{\text{d}}$ collects spring constants $\hat{w}_{x,i}^{\text{d}}$ of density-based spring connected to cell i . Hold force $\mathbf{f}_x^{\text{hold}}$ decouples successive placement iterations by canceling out net force that pulls cells back to the initial placement, and is defined as

$$\mathbf{f}_x^{\text{hold}} = -(\mathbf{C}_x \mathbf{x}' + \mathbf{d}_x), \quad (5)$$

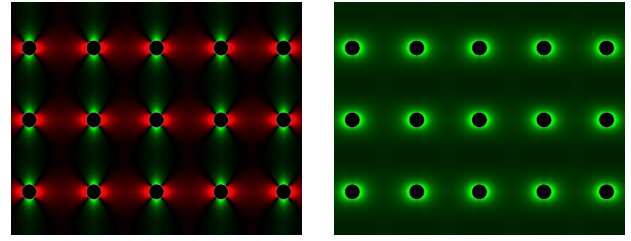
where vector $\mathbf{x}'$ represents the x-position of cells from last iteration. The placement result for each placement iteration can be obtained by setting total force $\mathbf{f}_x$ to zero, or solve

$$\mathbf{f}_x = \mathbf{f}_x^{\text{net}} + \mathbf{f}_x^{\text{den}} + \mathbf{f}_x^{\text{hold}} = \mathbf{0}. \quad (6)$$

B. Carrier Mobility-Based Forces

To consider the effect of TSV stress-induced carrier mobility variation during global placement, we need to introduce two additional forces, one for hole mobility variation $\mathbf{f}_x^{\text{mobil,h}}$ and another for electron mobility variation $\mathbf{f}_x^{\text{mobil,e}}$, into (6).

Here, $\mathbf{f}_x^{\text{mobil,h}}$ and $\mathbf{f}_x^{\text{mobil,e}}$ can be separately defined because they aim to optimize delay of different devices, e.g. PMOS and



(a) Hole mobility (b) Electron mobility
Fig. 3. Carrier mobility variation surface surrounding TSVs.

NMOS. For brevity, only description related to hole mobility is given in this paper because it similarly applies to electron mobility. The force can be represented by hole mobility-based springs connected to cells, and defined as

$$\mathbf{f}_x^{\text{mobil,h}} = \mathbf{C}_x^{\text{m,h}} (\mathbf{x} - \hat{\mathbf{x}}^{\text{m,h}}), \quad (7)$$

where vector $\hat{\mathbf{x}}^{\text{m,h}} = [\hat{x}_1^{\text{m,h}} \dots \hat{x}_N^{\text{m,h}}]^T$ represents the x-position of target points to which N cells are connected by hole mobility-based springs, and diagonal matrix $\mathbf{C}_x^{\text{m,h}}$ collects spring constants $\hat{w}_{x,i}^{\text{m,h}}$ of hole mobility-based spring connected to cell i .

Hole mobility-based target point $\hat{x}_i^{\text{m,h}}$ on die d is defined as

$$\hat{x}_i^{\text{m,h}} = x'_i + l_i^{\text{m,h}} \cdot \left. \frac{\frac{\partial}{\partial x} \Phi^{\text{m,h}}(x, y)}{\|\nabla \Phi^{\text{m,h}}(x, y)\|} \right|_{(x'_i, y'_i), z=d}, \quad (8)$$

where vector $\mathbf{x}' = [x'_1 \dots x'_N]^T$ represents the x-position of N cells from last iteration, $\Phi^{\text{m,h}}(x, y) = \frac{\Delta \mu}{\mu_{\text{total}}}(x, y)$ is hole mobility variation surface charted by using the model described in Section II, $\nabla \Phi^{\text{m,h}}(x, y)$ is its gradient, and $l_i^{\text{m,h}}$ is a length along the gradient direction of the surface. The gradient is added to current position in this equation because we want to move cell in the direction of hole mobility increase.

The carrier mobility variation surfaces are shown in Fig. 3. The green area in the figures indicates carrier mobility enhancement caused by TSV-induced stress, and the red area indicates degradation. Unlike placement density, carrier mobility variation surface is smooth (except at TSV edge because mobility variation is not defined inside TSVs). Therefore, we can determine the direction of target points to which cells are connected by mobility-based springs from its gradient directly.

1) *Balancing Forces*: The newly introduced $\mathbf{f}_x^{\text{mobil,h}}$ needs to be balanced against $\mathbf{f}_x^{\text{den}}$ and $\mathbf{f}_x^{\text{net}}$ (no need to balance against $\mathbf{f}_x^{\text{hold}}$). The force-directed quadratic placement in [10] already has a mechanism to balance $\mathbf{f}_x^{\text{den}}$ against $\mathbf{f}_x^{\text{net}}$ so that the speed of cell spreading is regulated across placement iterations. We can use the same mechanism, and, therefore, balance $\mathbf{f}_x^{\text{mobil,h}}$ against only $\mathbf{f}_x^{\text{den}}$. The parameters that need adjustment are the length along the gradient direction of hole mobility variation surface $l_i^{\text{m,h}}$ and hole mobility-based spring constant $\hat{w}_{x,i}^{\text{m,h}}$.

We choose $l_i^{\text{m,h}}$ so that hole mobility at the target point is higher than that at the current cell position. We start with

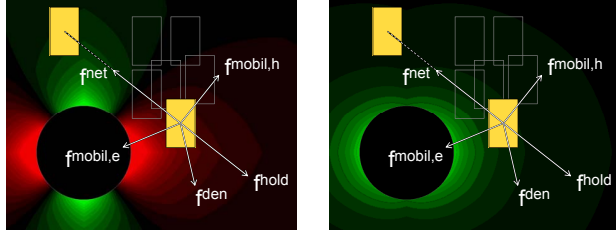


Fig. 4. All forces applied to a cell.

the length of $1/8 \times$ average cell size, and increase it to $1/4$, $1/2$, and $1 \times$ average cell size while hole mobility increases. We stop at the length of average cell size so that we do not increase wirelength too much. If hole mobility at even $1/8 \times$ average cell size is lower than that at the current cell position, we do not apply hole mobility-based force to that cell at all in that iteration.

Compared to density-based gradient, which directly defines the length to density-based target point in (4), $l_i^{m,h}$ is relatively constant. Density-based gradient is extremely high in early placement iterations because of cell overlap, and decreases to almost zero as overlap is resolved in late iterations [10]. By limiting $l_i^{m,h}$ to average cell size, we naturally balance it to the length to density-based target point. The length to density-based target point dominates during early iterations, and the effect of $l_i^{m,h}$ becomes pronounced when the length to density-based target point drops below $l_i^{m,h}$ during late iterations.

During global placement, we perform SA 3D STA periodically. The results from SA 3D STA include the set of cells whose rise- and/or fall-time slack is negative. Then, we balance hole mobility-based spring constant $\hat{w}_{x,i}^{m,h}$ against density-based spring constant $\hat{w}_{x,i}^d$ of diagonal matrix $\hat{\mathbf{C}}_x^d$ in (4) by defining it as

$$\hat{w}_{x,i}^{m,h} = c_i^{h,j} \times \hat{w}_{x,i}^d, \quad (9)$$

where $c_i^{h,j}$ is rise-time criticality of cell i after j^{th} SA 3D STA, and defined as

$$c_i^{h,j} = \begin{cases} (c_i^{h,j-1} + s_i^{h,j}/S_{\min}^j)/2 & \text{if } i \in \mathbb{C}_c^{h,j} \\ c_i^{h,j-1}/2 & \text{otherwise,} \end{cases} \quad (10)$$

where $s_i^{h,j}$ is rise-time slack of cell i , $S_{\min}^j$ is minimum timing slack of the design, and $\mathbb{C}_c^{h,j}$ is the set of cells whose rise-time slack is negative and less than 90% of $S_{\min}^j$. In other words, we determine cell rise-time criticality based on its history and current rise-time slack. Therefore, the effect of hole mobility-based spring is pronounced on a cell whose rise-time is highly critical, and needs hole mobility enhancement.

2) *New Total Force*: An illustration of all forces applied to a cell is shown in Fig. 4. In the figure, $\mathbf{f}^{\text{net}}$ tries to hold the yellow cells of a net together, but $\mathbf{f}^{\text{hold}}$ tries to nullify its effect, allowing cells to be moved based on other forces. Because of high cell density on top of right yellow cell, $\mathbf{f}^{\text{den}}$

TABLE I
BENCHMARK CIRCUITS.

Circuit	#Gates	#Nets	#TSVs	Profile
ckt1	20K	20K	634	Microprocessor
ckt2	33K	33K	3,554	Arithmetic Unit
ckt3	50K	51K	5,352	Connection Bus
ckt4	80K	80K	2,846	Network Controller
ckt5	119K	119K	5,341	Data Encryption

tries to move the cell down. If the cell is rise-time critical, $\mathbf{f}^{\text{mobil,h}}$ tries to move the cell toward top right, away from the TSV, where hole mobility degradation decreases as shown in Fig. 4(a). If the cell is, however, fall-time critical, $\mathbf{f}^{\text{mobil,e}}$ tries to move the cell left, toward the TSV, where electron mobility increases as shown in Fig. 4(b). In the case that a cell is both rise- and fall-time critical, the result depends on which timing is more critical.

With the newly introduced hole mobility-based force $\mathbf{f}_x^{\text{mobil,h}}$ and electron mobility-based force $\mathbf{f}_x^{\text{mobil,e}}$, the total force becomes

$$\mathbf{f}_x = \mathbf{f}_x^{\text{net}} + \mathbf{f}_x^{\text{hold}} + \mathbf{f}_x^{\text{den}} + \mathbf{f}_x^{\text{mobil,h}} + \mathbf{f}_x^{\text{mobil,e}}. \quad (11)$$

By setting $\mathbf{f}_x = \mathbf{0}$ and substituting equations, the new result for each placement iteration can be obtained by solving

$$(\mathbf{C}_x + \hat{\mathbf{C}}_x^d + \hat{\mathbf{C}}_x^{m,h} + \hat{\mathbf{C}}_x^{m,e})\Delta\mathbf{x} = -\hat{\mathbf{C}}_x^d\Phi^d + \hat{\mathbf{C}}_x^{m,h}\Phi^{m,h} + \hat{\mathbf{C}}_x^{m,e}\Phi^{m,e} \quad (12)$$

for $\Delta\mathbf{x}$, where vector $\Delta\mathbf{x} = \mathbf{x} - \mathbf{x}'$ indicates how far cells should be moved, Φ^d is the vector collecting density-based gradients, and $\Phi^{m,h}$ and $\Phi^{m,e}$ are the vectors collecting $l_i^{m,h} \cdot \frac{\partial \Phi^{m,h}}{\partial x} / \|\nabla \Phi^{m,h}\|$ and $l_i^{m,e} \cdot \frac{\partial \Phi^{m,e}}{\partial x} / \|\nabla \Phi^{m,e}\|$ from (8).

C. Convergence of Stress-Driven Global Placement

We found that introducing $\mathbf{f}^{\text{mobil,h}}$ and $\mathbf{f}^{\text{mobil,e}}$ to 3D force-directed quadratic placement without proper monitoring may cause problem to its convergence. During the early iterations of designs with irregular TSV position, highly overlapping TSVs in a region result in extremely high mobility variation, which can misguide the placer. Because TSVs are also moved in every placement iteration as well to resolve their overlap, carrier mobility surfaces change. Critical cells are pulled by overlapping TSVs, worsening wirelength, until the overlap is finally resolved, and realize that the mobility improvement already vanishes. To prevent this problem, we put an upper-bound limit on mobility variation from (1).

Another problem arises when a cell is moved over the top of a TSV or its KOZ during placement iterations. When a cell is inside a TSV, (1) is not defined. Also the mobility variation is not valid when a cell is inside KOZ because the cell is moved out of KOZ during legalization. We do not apply $\mathbf{f}^{\text{mobil,h}}$ and $\mathbf{f}^{\text{mobil,e}}$ to the cell under these cases to prevent the placer from being misguided.

V. EXPERIMENTAL RESULTS

We use IWLS 2005 benchmarks [11] and several industrial circuits as listed in Table I. We use 45nm technology for

TABLE II
IMPACT OF KOZ ON CARRIER MOBILITY VARIATION FOR CKT5.

TSV Cell	Mobility Variation (%)			
	Regular TSV Position		Irregular TSV Position	
	Hole	Electron	Hole	Electron
2-row	-4.56 – 2.81	0.33 – 3.62	-8.76 – 4.74	0.02 – 7.47
3-row	-4.05 – 2.45	0.30 – 2.35	-6.04 – 2.55	0.02 – 4.10
4-row	-2.07 – 1.51	0.26 – 1.37	-2.87 – 1.75	0.02 – 2.17
5-row	-1.55 – 0.93	0.18 – 0.89	-2.30 – 1.26	0.02 – 1.32
6-row	-1.01 – 0.70	0.13 – 0.64	-1.33 – 0.70	0.02 – 0.85
7-row	-0.90 – 0.53	0.07 – 0.40	-1.15 – 0.61	0.02 – 0.62

TABLE III
IMPACT OF KOZ ON AREA AND WIRELENGTH FOR CKT5.

TSV Cell	Footprint (mm ²)	TSV Cell Area (mm ²)	Wirelength (m)	
			Regular	Irregular
			TSV Position	TSV Position
2-row	0.176 (1.00)	0.130 (18.47%)	3.415 (1.00)	2.970 (1.00)
3-row	0.250 (1.42)	0.293 (29.33%)	3.970 (1.16)	3.475 (1.17)
4-row	0.360 (2.04)	0.521 (36.21%)	4.726 (1.38)	4.196 (1.41)
5-row	0.504 (2.86)	0.815 (40.40%)	5.526 (1.62)	4.654 (1.57)
6-row	0.672 (3.81)	1.173 (43.61%)	6.331 (1.85)	5.328 (1.79)
7-row	0.884 (5.01)	1.597 (45.17%)	7.179 (2.10)	6.036 (2.03)

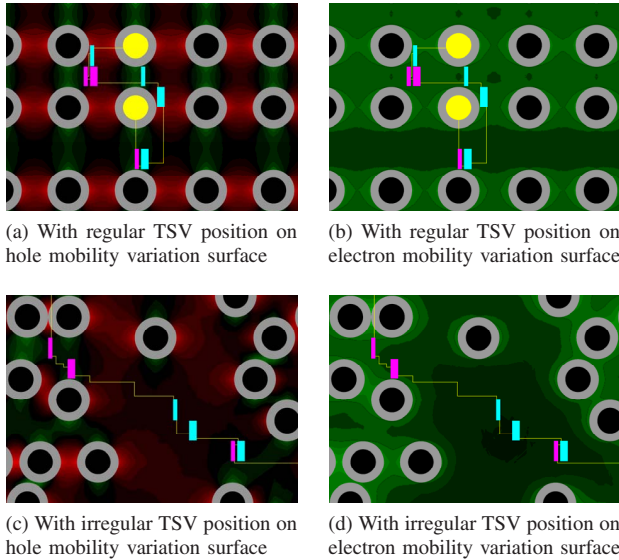


Fig. 5. Zoom-up snapshots of stress-driven placement results for ckt3 using 2-row TSV cells.

our experiments with TSV size of $3\mu\text{m}$. The TSV parasitic capacitance and resistance are 50fF and 0.2Ω , respectively. We expand KOZ around TSVs to make TSV cells (= TSV + KOZ) fit inside 2–7 standard-cell rows (1 standard-cell row = $2.47\mu\text{m}$). We base all our experiments on 4-die chip stack with constant cell area density. We use min-cut partitioner, and set the target clock period to the value reported after synthesis. All reported timing results come from SA 3D STA.

A. Impact of KOZ on carrier mobility variation

In this experiment, we increase TSV cell size from 2-row to 7-row while observing carrier mobility variation caused by TSV-induced stress. The results are shown in Table II. The result indicates that carrier mobility variation decreases as

KOZ increases, and starts becoming negligible (1% or less) when TSV cell size reaches 6-row. We also observe that mobility variation in design with irregular TSV position is larger than that in design with regular TSV position. TSV cells in design with irregular TSV position can be crowded in some area, causing high stress and mobility variation.

B. Impact of KOZ on area and wirelength

The main purpose of KOZ is to prevent cells from being placed so close to TSV that they experience carrier mobility variation. The side effect of enforcing large KOZ to have predictable device performance is shown in Table III. The footprint area of the chip stack for ckt5 increases almost $4\times$ if TSV cell size is 6-row. The increased area is primarily consumed by TSV cells. In an extreme case, almost half of silicon area is consumed by TSV cells when TSV cell size is 7-row. Increasing footprint area inevitably results in $2\times$ wirelength increase due to large KOZ choice.

C. Impact of KOZ on stress-aware timing

We run SA 3D STA after obtaining placement results from wirelength-driven, timing-driven, and stress-driven placers. The results are shown in Table IV. First, we observe that, under TSV-induced stress, the timing results from timing-driven placement can be unpredictable, and worse than the results from even wirelength-driven placement in many cases. Traditional timing-driven placer is oblivious to the change in carrier mobility of devices, and only tries to reduce the capacitive load on timing critical cells. Second, we observe that our stress-driven placer outperforms timing-driven placer consistently. The improvements over wirelength-driven placement on worst negative slack (WNS) and total negative slack (TNS) are up to 39% and 42% respectively. Third, we also observe that, using 2-row TSV cells, our stress-driven placer provides better result for design with irregular TSV position than the result for design with regular TSV position. Design with irregular TSV position has shorter wirelength and higher carrier mobility variation which our placement algorithm can intelligently exploit. Finally, we observe that, as TSV cell size increases, the benefit from our stress-driven placement decreases. Large KOZ leaves not much mobility variation for our stress-driven placement to exploit.

D. Stress-driven placement results

We obtain placement results from our stress-driven placer. The snapshots of ckt3 are shown in Fig. 5. In the figures, gray band surrounding TSVs is KOZ. Logic cells in magenta are hole mobility critical cells. Their timing arcs are rising on the critical paths. Our placer positions them (if possible) in green area of Fig. 5(a) where they receive hole mobility enhancement, or, at least, in black area where they do not experience hole mobility degradation. On the other hand, logic cells in sky blue are electron mobility critical cells. Their timing arcs are falling on the critical paths. Our placer positions them (if possible) in bright green area of Fig. 5(b) where they receive higher electron mobility enhancement.

TABLE IV
IMPACT OF KOZ ON STRESS-AWARE TIMING FOR CKT5.

TSV Cell	Regular TSV Position						Irregular TSV Position					
	Wirelength-Driven		Timing-Driven		Stress-Driven		Wirelength-Driven		Timing-Driven		Stress-Driven	
	WNS (ps) (=100%)	TNS (ps) (=100%)	WNS (%)	TNS (%)	WNS (%)	TNS (%)	WNS (ps) (=100%)	TNS (ps) (=100%)	WNS (%)	TNS (%)	WNS (%)	TNS (%)
2-row	-92.72	-143	113.66	126.57	77.01	69.93	-79.26	-120	127.26	143.33	60.62	57.50
3-row	-96.62	-156	70.14	65.38	70.60	62.82	-77.89	-118	133.11	147.46	94.12	92.37
4-row	-102.86	-170	85.06	82.94	78.20	74.71	-85.42	-134	111.53	114.18	92.95	90.30
5-row	-99.28	-157	88.43	87.90	88.48	87.90	-88.32	-139	100.83	100.72	99.91	99.28
6-row	-88.45	-139	99.27	99.28	99.31	99.28	-88.33	-139	99.43	99.28	99.54	99.28
7-row	-88.55	-139	99.02	99.28	99.09	99.28	-88.28	-139	99.63	99.28	99.43	99.28

TABLE V
TIMING COMPARISON FOR REGULAR AND IRREGULAR TSV POSITION WITH 2-ROW TSVs.

Circuit	Regular TSV Position						Irregular TSV Position					
	Wirelength-Driven		Timing-Driven		Stress-Driven		Wirelength-Driven		Timing-Driven		Stress-Driven	
	WNS (ps)	TNS (ps)	WNS (ps)	TNS (ps)	WNS (ps)	TNS (ps)	WNS (ps)	TNS (ps)	WNS (ps)	TNS (ps)	WNS (ps)	TNS (ps)
ckt1	-163.50	-1,167	-156.67	-1,034	-156.39	-1,034	-157.04	-1,063	-161.21	-1,107	-155.15	-1,004
ckt2	-159.35	-5,104	-180.86	-6,076	-129.35	-4,105	-127.28	-4,005	-134.70	-4,327	-120.45	-3,888
ckt3	-79.35	-605	-65.85	-428	-53.72	-321	-73.40	-482	-56.65	-348	-51.88	-307
ckt4	-55.39	-131	-49.25	-106	-38.01	-72	-50.38	-109	-40.75	-80	-34.95	-66
ckt5	-92.72	-143	-105.39	-181	-71.40	-100	-79.26	-120	-100.87	-172	-48.05	-69
AVE (%)	100.00	100.00	98.98	97.18	78.03	69.40	100.00	100.00	98.76	100.22	78.82	74.65

The results from different placement algorithms using 2-row TSV cells are shown in Table V. On average, timing-driven placer does not provide performance improvement over wirelength-driven placer when evaluated by SA 3D STA. The cells on critical paths may be placed in location that carrier mobility is degraded by TSV-induced stress. On the other hand, stress-driven placer consistently provides better performance than the other placers. On average, the performance improvement over wirelength-driven placement on WNS and TNS are 21.6% and 28.0% respectively. We observe again that the results for design with irregular TSV position are better than those for design with regular TSV position in all cases.

VI. CONCLUSIONS

In this paper, we analyzed the impact of choice of KOZ dimension around TSVs on the mechanical stress, carrier mobility variation, area, wirelength, and performance of 3D ICs. Use of large KOZs was observed to practically nullify the impact of TSV-induced stress on carrier mobility but at the cost of almost $4\times$ increase in chip stack footprint area and $2\times$ increase in wirelength. In addition, we observed that, to regain footprint area by reducing KOZ dimension, 3D-IC placers must be TSV stress-aware to extract best circuit performance by *exploiting* the TSV stress-dependent mobility variation instead of avoiding it. Lastly, we proposed the first TSV-induced stress-driven force-directed 3D global placer by adding new carrier (hole and electron) mobility-based forces to traditional force-directed placement. Compared to wirelength-driven placement, our placer results in superior placement with average reductions of WNS by 21.6% and TNS by 28.0%.

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